

$$A = \begin{pmatrix} -1 & -2 & 0 \\ -1 & 3 & -1 \\ 2 & 3 & 1 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} -1-\lambda & -2 & 0 \\ -1 & 3-\lambda & -1 \\ 2 & 3 & 1-\lambda \end{pmatrix}$$

$$\det(A - \lambda I) = -(\lambda + 1)(\lambda - 2)^2$$

$$\lambda_1 = -1 \quad \lambda_2 = 2 \text{ (multiplicity 2)}$$

$$\lambda_1 = -1 \quad (A + I)x = 0$$

$$\begin{pmatrix} 0 & -2 & 0 \\ -1 & 4 & -1 \\ 2 & 3 & 2 \end{pmatrix} \quad v_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \quad \text{Gen. 1}$$

$$\lambda_2 = 2 \quad (A - 2I)x = 0$$

$$\begin{pmatrix} -3 & -2 & 0 \\ -1 & 1 & -1 \\ 2 & 3 & -1 \end{pmatrix} \quad \vec{v}_2 = \begin{pmatrix} 2 \\ -3 \\ -5 \end{pmatrix} \quad \text{Gen. 1}$$

⤴
Hledáme v_3 :

$$(A - 2I)\vec{v}_3 = \vec{v}_2$$

$$\vec{v}_3 = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix}$$

$$U_1 = \text{span} \{ \vec{v}_1, \vec{v}_2 \} \quad B_{U_1} = (\vec{v}_1, \vec{v}_2)$$

$$[f_1 | v_1]_{B_{v_1}}^{B_{v_1}} = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$$

$$U_2 = \text{Span} \{ \vec{v}_2, \vec{v}_3 \} \quad B_{U_2} = (\vec{v}_2, \vec{v}_3)$$

$$[f_1 | v_2]_{B_{U_2}}^{B_{U_2}} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$